

A1H's pinkish theme attracts attention. The peach board is a significant step in advancement over the earlier (~100 Mhz), Kaede and Sakura boards. The board working up to 400 MHz embeds an ARM Cortex-A9 central processing unit (CPU) core to handle the processing. The RZ/A1H 324 pin BGA includes a 32 kB L1 instruction cache, another 32 kB L1 data cache along with a 128 kB L2 cache.

Peripherals on the peach board

Among the peripherals integrated on the board, for data input we have universal serial bus (USB) and the ethernet interface. These have become the most common modes of data transfer in wired means of communication. Other interfaces include the serial parallel interface (SPI), inter integrated circuit (I2C) and the universal asynchronous receiver transmitter (UART).

Using the board?

The forum explains some of the use cases with a camera, and employing the board as a web cameras. A JPEG codec unit is additionally provided on the board. Two image display sized Wide extended graphics array (WXGA) can be enabled just by the 10 MB onboard random access memory (RAM). This seems to be a significant development compared to the earlier boards, which required external synchronous dynamic RAM (SDRAM) to extend the capabilities resulting in a bigger device.

Renesas went a step ahead with power consumption this time and the input power is rated at 3.3V. The ARM Cortex CPU however works at 1.8 V. The different power modes, further optimise operation of the board. The device can be booted in several different modes, along with the SD memory connected to the system.

The Arduino UNO pin compatibility, supports arduino shields that are easy to mount and cheap to produce. Shields are boards that when added to a development board, extend the original board capabilities. Some of the interesting shields to use with the Peach board would be the Arduino Xbee, for wireless connectivity in case your USB ports are busy, and the various motor shields for a small home automation system



- ▶ 10 MB RAM
- ▶ 2x USB host/device interface, 1x ethernet interface
- ▶ 3x SPI, 3x I2C
- ▶ 8x UART
- ▶ 7x12bit ADC
- ▶ 2x camera input
- ▶ JPEG codec, video decoder
- ▶ Open VG1.1 2D graphics accelerator
- ▶ WiFi (BP3595 module via USBHost)

Check-out some features

Memory comes as separate 32 kB partitions for instruction and data each. Additionally 128 kB secondary level cache with CoreLink Level 2 cache controller L2C-310 comes in. Apart from the 10 MB RAM, the board supports 384 MB(64 MB x 6) addressable external memory. The 8/16/32 bit data bus clocked at a maximum of 66.67 MHz suits well with the system.

Equipped with five multifunction 16-bit timers, the board consists of a real time clock along with 8 channels for motor control pulse width modulation (PWM) timer.

The board provides ample solutions for connectivity. The NAND flash interface supported by the two USB 2.0 host/function modules, should be the most used port. Two channels for SD card and the ethernet controller at 10/100 Mbps seem suitable. Asyn-

chronous and clock synchronous serial communication is made possible via the 8 channels of serial communication interface with 16-stage FIFO.

Additionally the board consists of two local interconnect network (LIN) interface, four I2C, five controller area network (CAN) along with five Renesas serial peripheral interface.

Employing the board

The GR-PEACH looks like a solution for industry usage and can also be employed as car accessory. The two channels for smartcard IrDA 1.0 interface back it up. These find suitable use at company gateways. The encryption engine provides additional support.

To put the board to practical use, we require an IDE to input the drivers. At present, we have the mbed online compiler. IDE for GR is still tentative and expected to feature support for the Peach board soon. Proper support in the IDE for GR, would enable programming it offline.

After installing a driver for serial configuration (also available at the mbed site), via a USB connection between the board and your PC, the board is ready to use. You can import pre-written programs to the board, directly via the computer interface. The program runs on pressing the peach's reset button. Other features of the board like, A/D converter, SPI, MicroSD, ethernet, I2C can be used in a similar manner after making minor adjustments in the provided code.

Get set, go

The board is available in two versions. The GR-PEACH embeds ARM Cortex-A RZ/A1H (R7S721001VCBG). Whereas the GR-PEACH FULL board comes equipped with local area network (LAN) connector (RJ-45) and connectors for Arduino UNO compatible pins and service pins.

The general specifications of the board make it suitable for the target market. The specifications look enough to get the job done efficiently. With it being a development board, also adds the opportunity for you to get your hands dirty and get to some tinkering. So get going.

Renesas is also coming out with a design contest related to this board. To make sure you do not miss the announcement, please visit <http://gadget.renesas.com/en/product/peach.html>